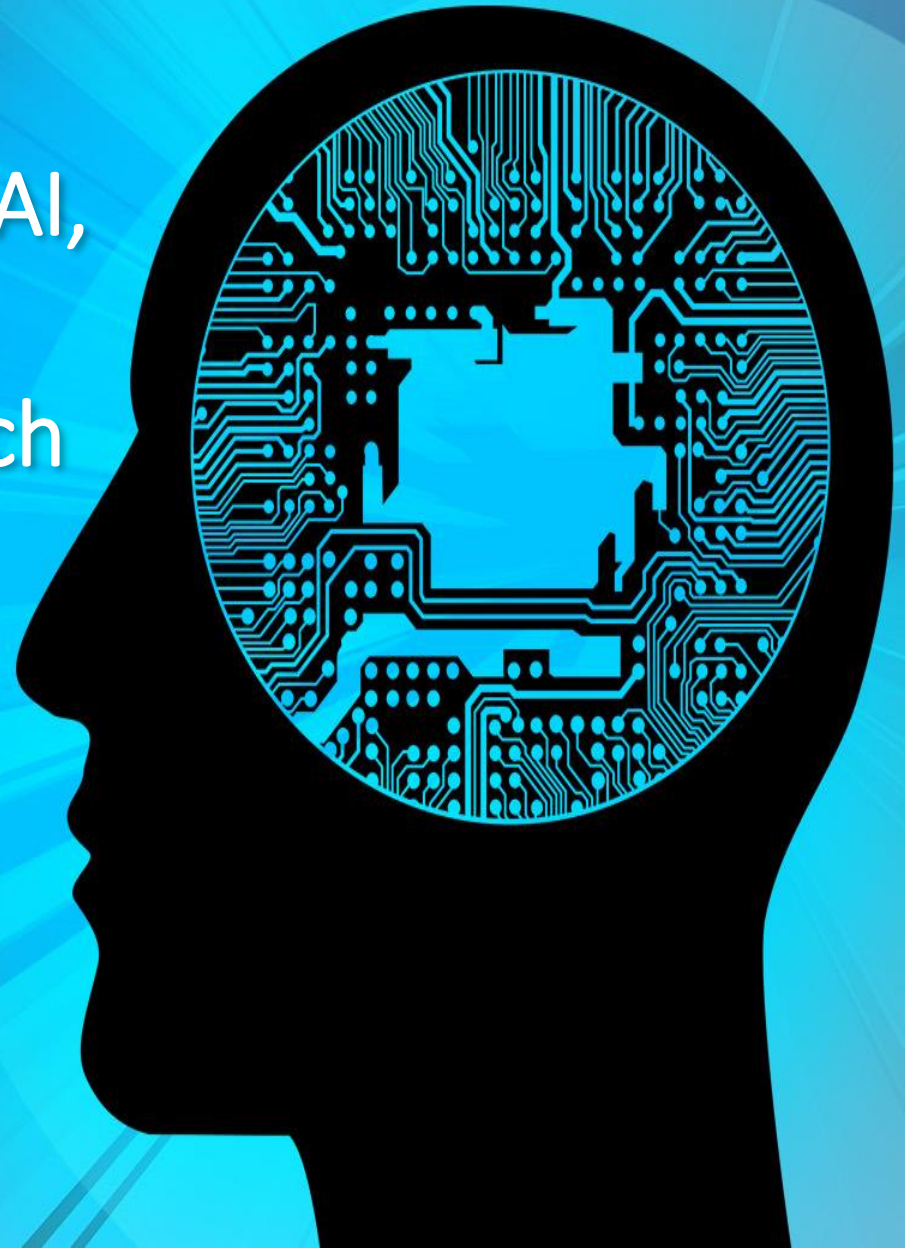


Past and Present Experiences with Knowledge Graphs for AI, Cognitive Science and Human Factors Research

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KNOWLEDGE GRAPHS

- A **knowledge graph** is a graph-based data model for representing real-world knowledge, where entities are modelled as labelled nodes, and relationships as labelled edges.
- Many iterations (and variations) in AI research
- **KNOWBEL** – my first project in this area – integration of object-oriented knowledge base with symbolic reasoning for AI
 - Initially applied to modelling nuclear reactors with automated alarm filtering using AI – (impossible with relational databases) – U. of T. AI

AI-Based Knowledge Representations to Represent Complex Domains

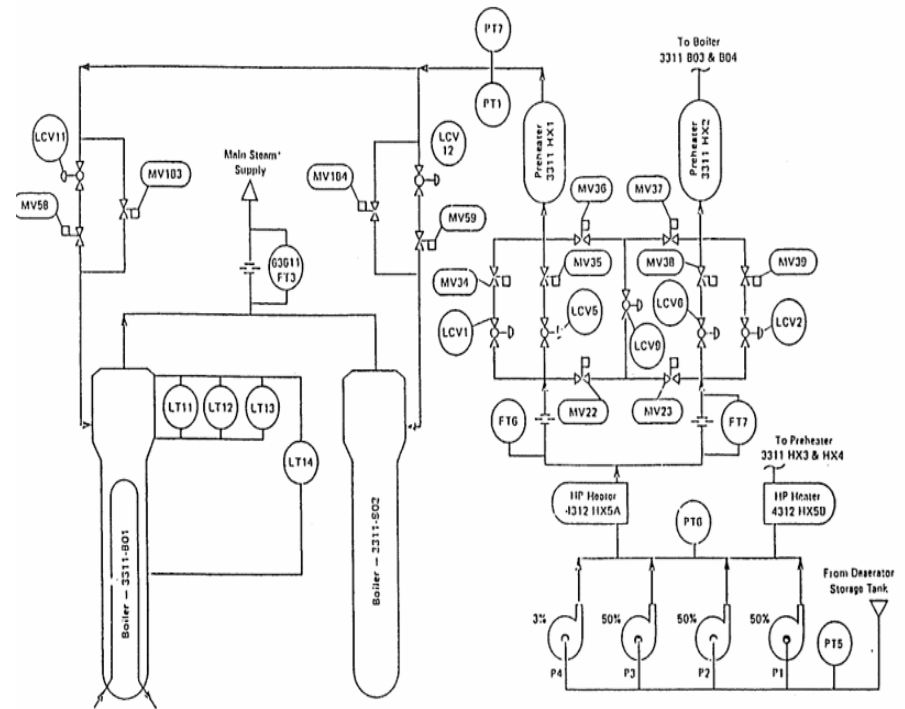
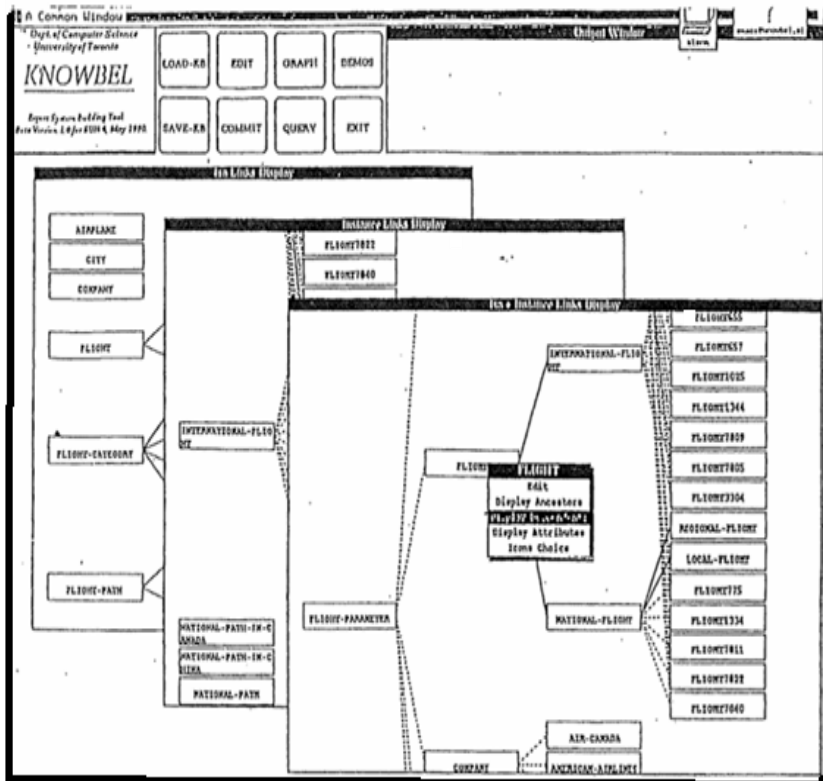
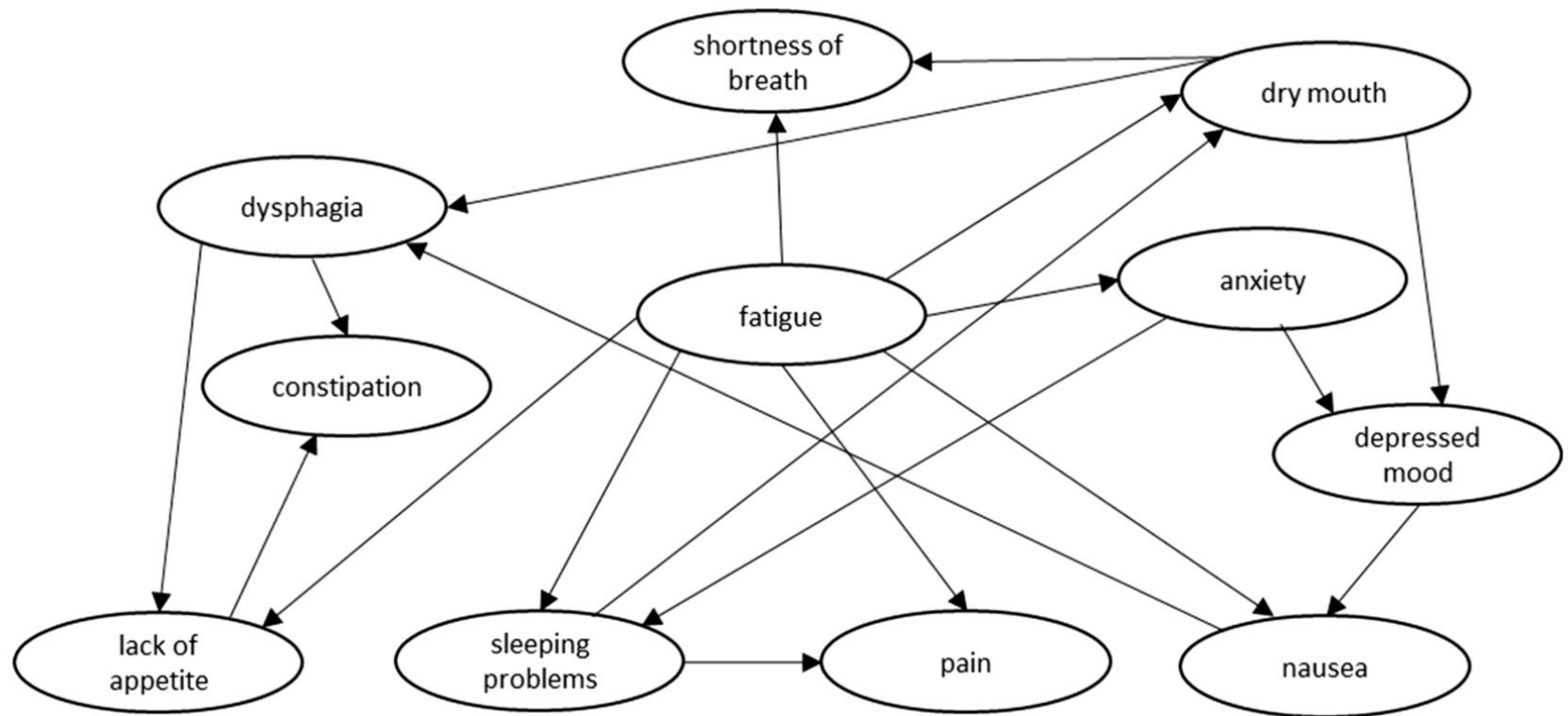


Figure 2: A Simplified Schematic for a Boiler Feedwater System

BAYESIAN BELIEF NETWORKS

- **Bayesian Belief Networks (BBNs)**
 - probabilistic graphical models that represent variables and their conditional dependencies using a Directed Acyclic Graph
- Used for **knowledge representation** and clinical decision support (CDSS)
 - **Bayesian Belief Networks (BBNs)** -- a specialized type of knowledge graph, referred to as a **probabilistic knowledge graph**.
 - Graphical models used to represent entities, concepts, and the statistical relationships among them
- Used in **Artificial Intelligence** and **Machine Learning** to:
 - make predictions, perform risk analysis, and diagnose causes when dealing with uncertainty and incomplete information

BELIEF NET APPLICATIONS IN HEALTHCARE



APPLICATION TO STUDY OF MEDICAL COGNITION

- Work in **medical cognition** (at McGill U.) and introduced to semantic nets to represent and reason about human cognition.
- **Sematic nets** (Sowa) -- graph structures for representing knowledge in patterns of interconnected nodes and arcs. Earliest versions were used in philosophy, psychology and linguistics before being used for AI and Cognitive Science

Kushniruk AW, Patel VL, Marley AA. Small worlds and medical expertise: implications for medical cognition and knowledge engineering. International journal of medical informatics. 1998 May 1;49(3):255-71.

RESEARCH SUPPORTING THE “SMALL WORLDS” HYPOTHESIS

- **Research Question** – How do expert physicians make efficient diagnoses?
- And how can that knowledge be incorporated in an AI system (using a Bayesian belief net) to do diagnostic reasoning?
- **Small Worlds Hypothesis:** Experts develop small sets of closely related possible disease diagnoses and ask for information about **key distinguishing features** to narrow down, in a **step-wise process**

INFORMATION PROCESSING WITH SMALL WORLDS – SPECIFIC PREDICTIONS

- When an **expert** attempts to arrive at a diagnosis the physician is hypothesized as focusing on a small set of logically related diseases, i.e. small worlds, carrying out a limited number of comparisons.
- On the other hand it is hypothesized less than expert physicians (novice and intermediates) will generate large sets of diverse hypotheses
- Comparison of AI system to human processing supported the small worlds hypothesis
- Implications for both AI and medical education

Knowledge Graphs in AI research for Medical Cognition – Proving The “Small Worlds” Hypothesis

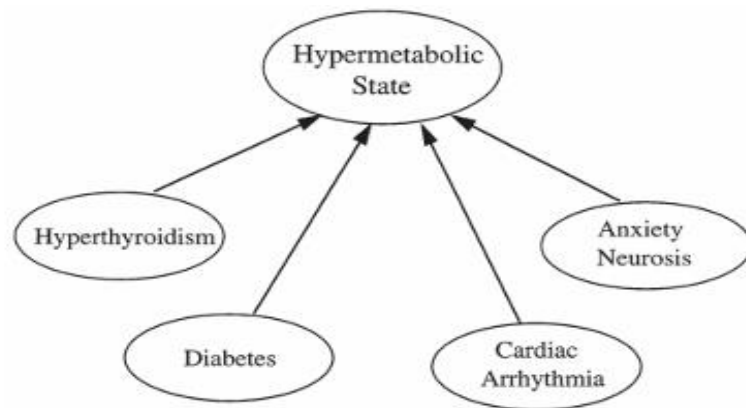


Fig. 1. Cluster of four related causes of a hypermetabolic state.

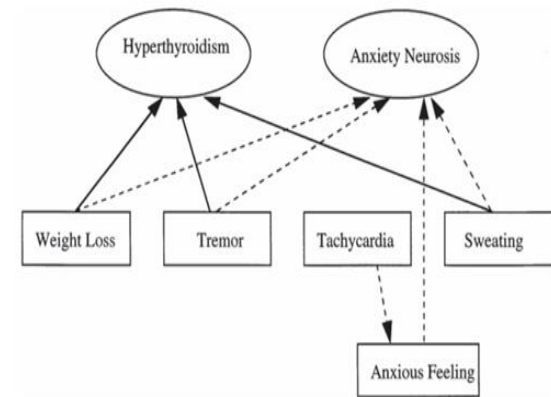
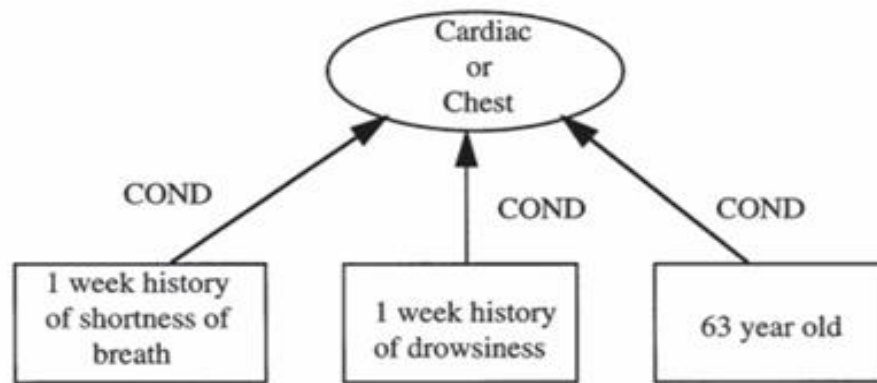


Fig. 2. Local knowledge map for distinguishing between hyperthyroidism and anxiety neurosis. Ovals contain diagnostic hypotheses and boxes contain findings. Arcs indicate probabilistic relations among findings and hypotheses.

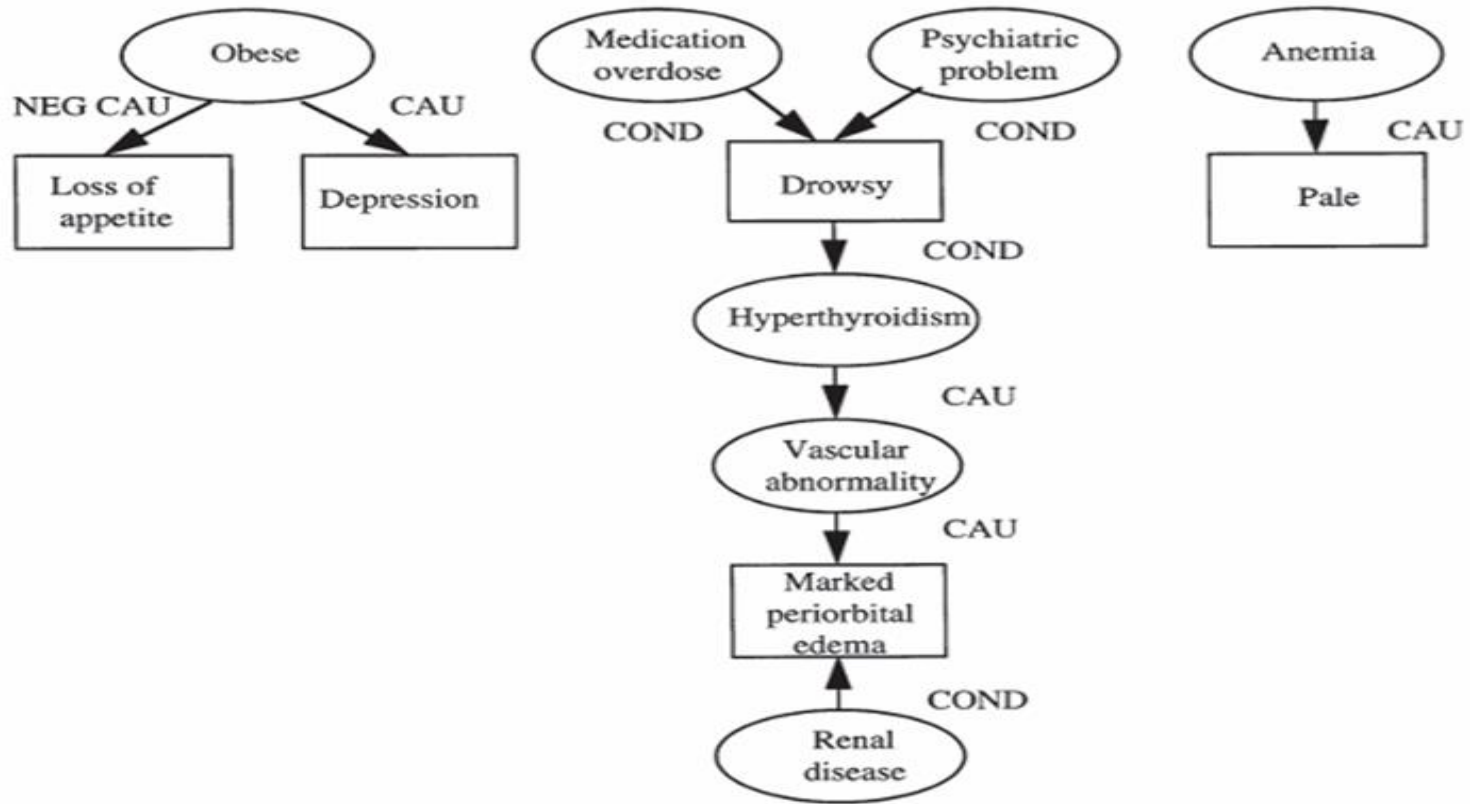


HDK Subject (expert)



LDK Subject (non-expert)

Fig. 4. Semantic network representation of verbal protocols produced by the expert HDK and non-expert LDK subjects in response to the first statement of an endocrine problem. Boxes contain the cues that the subject considered and ovals contain hypotheses that were entertained. COND indicates a conditional relation, while CAU indicates a



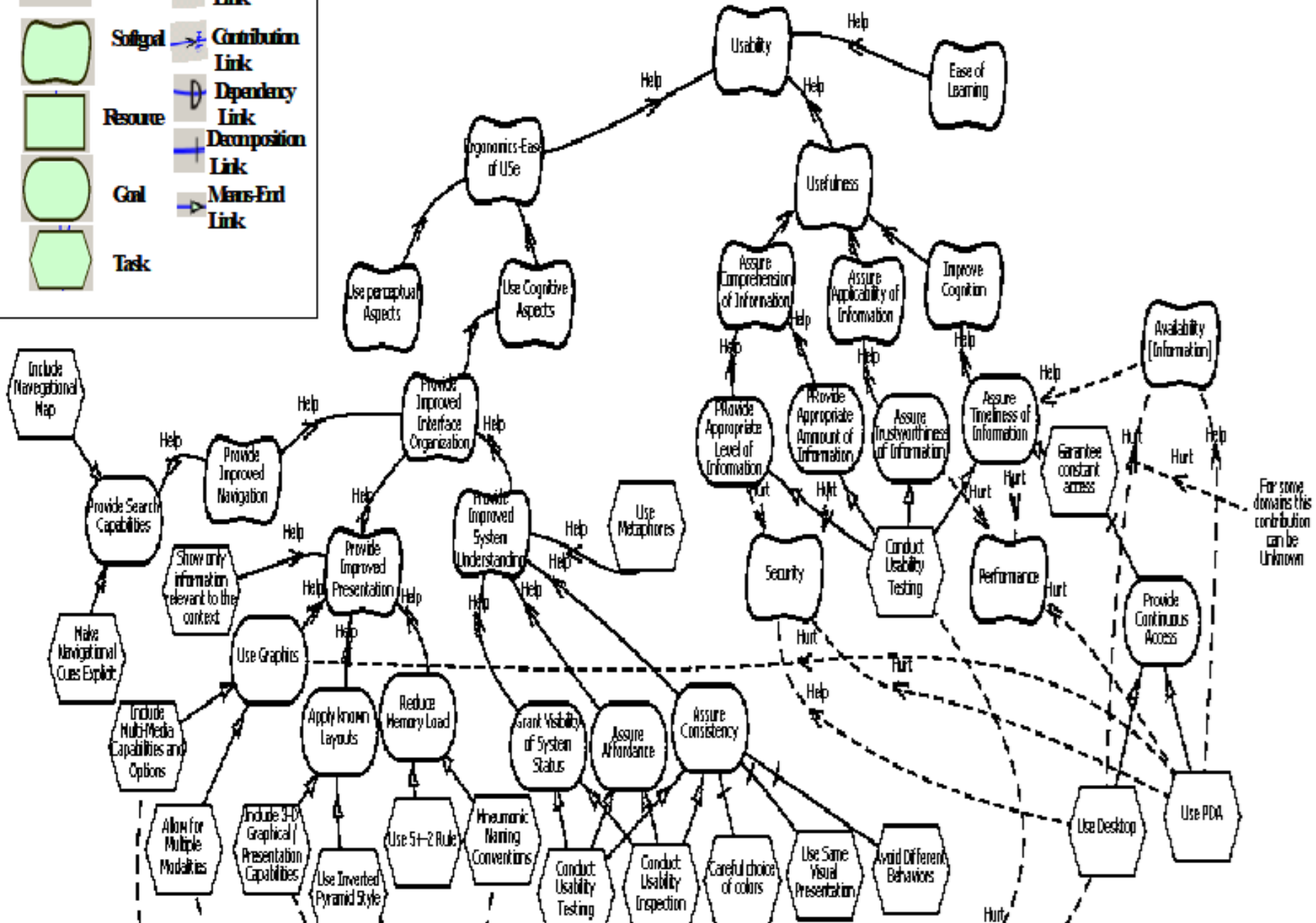
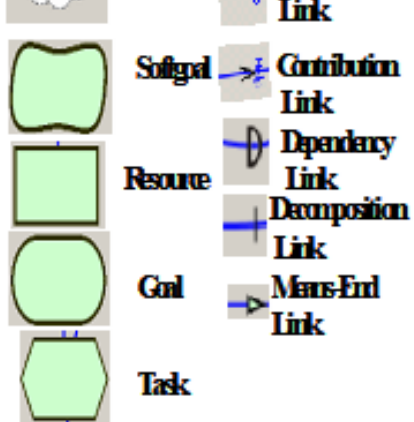
IMPLICATIONS

- Approach to reasoning from small worlds applied in **PATHFINDER AI system**
 - Designed to assist physicians in diagnosing complex **lymph-node diseases**, which is historically one of the most difficult areas of surgical pathology
 - **Probabilistic reasoning**
 - **Hypothesis-directed problem solving**
- Implications for both AI and for medical cognition and study of medical expertise
 - Example of **cognitive emulation** – working of system mirrored expert human decision making
 - Included explicit mapping of symptoms to diagnoses
 - AI system reasoning validated through cognitive studies (i.e. matched data from **think-aloud studies** of **human** problem solving)

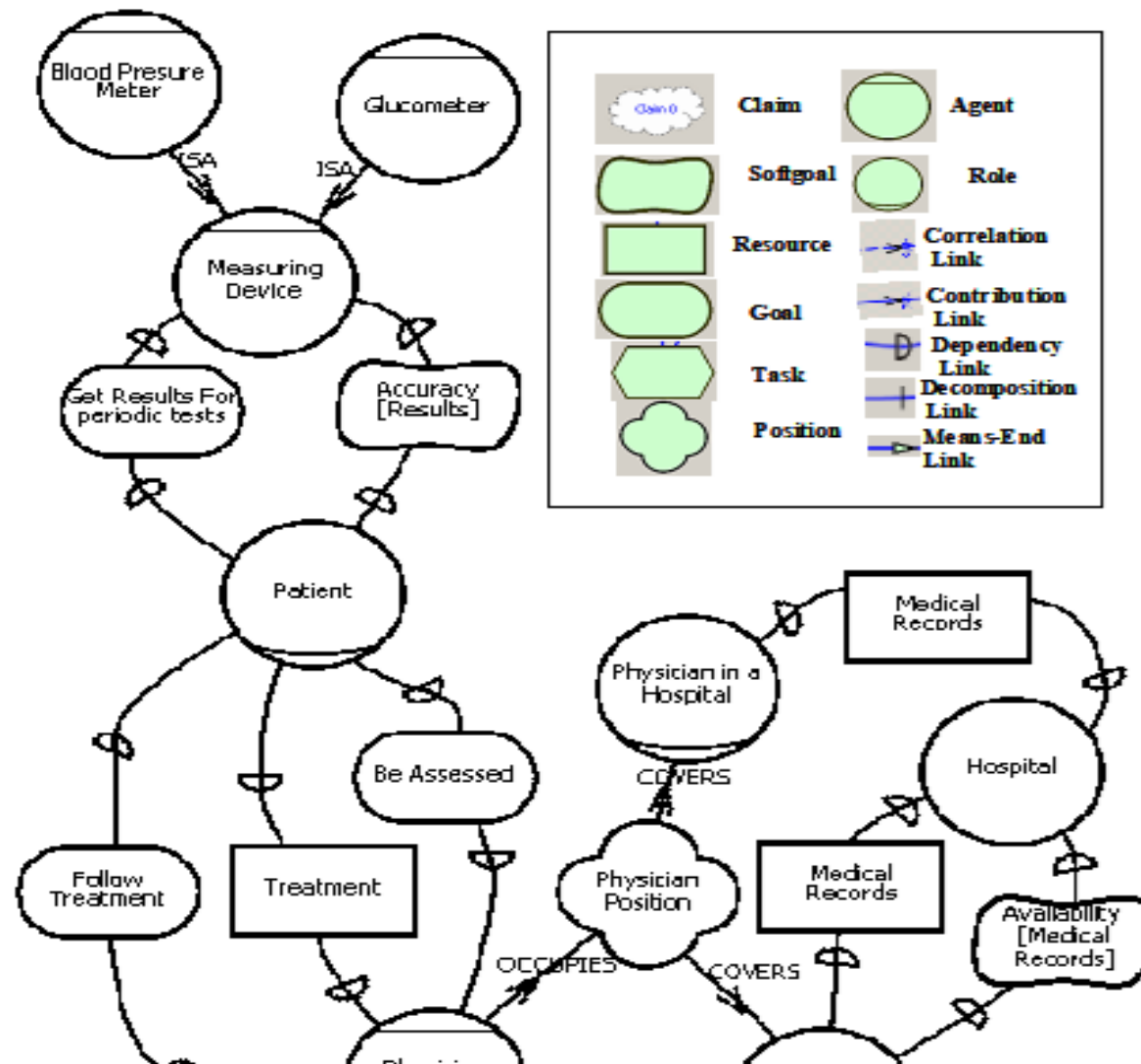
KNOWLEDGE GRAPHS FOR USE IN SPECIFICATION OF USABILITY REQUIREMENTS

- Development of a usability ontology, or classification of aspects of usability that can aid in the specification of usability requirements
- Use of i* modelling tool – an agent-oriented modelling approach used in requirements modelling
- Knowledge from various sources represented in a graph structure
- Links between concepts can represent dependencies, “helping” and “hurting”
- Can express different alternatives for achieving a goal (e.g. improved usability)

Cysneiros LM, Kushniruk A. Bringing usability to the early stages of software development. In Proceedings. 11th IEEE International Requirements Engineering Conference, 2003. 2003 Sep 12 (pp. 359-360). IEEE.

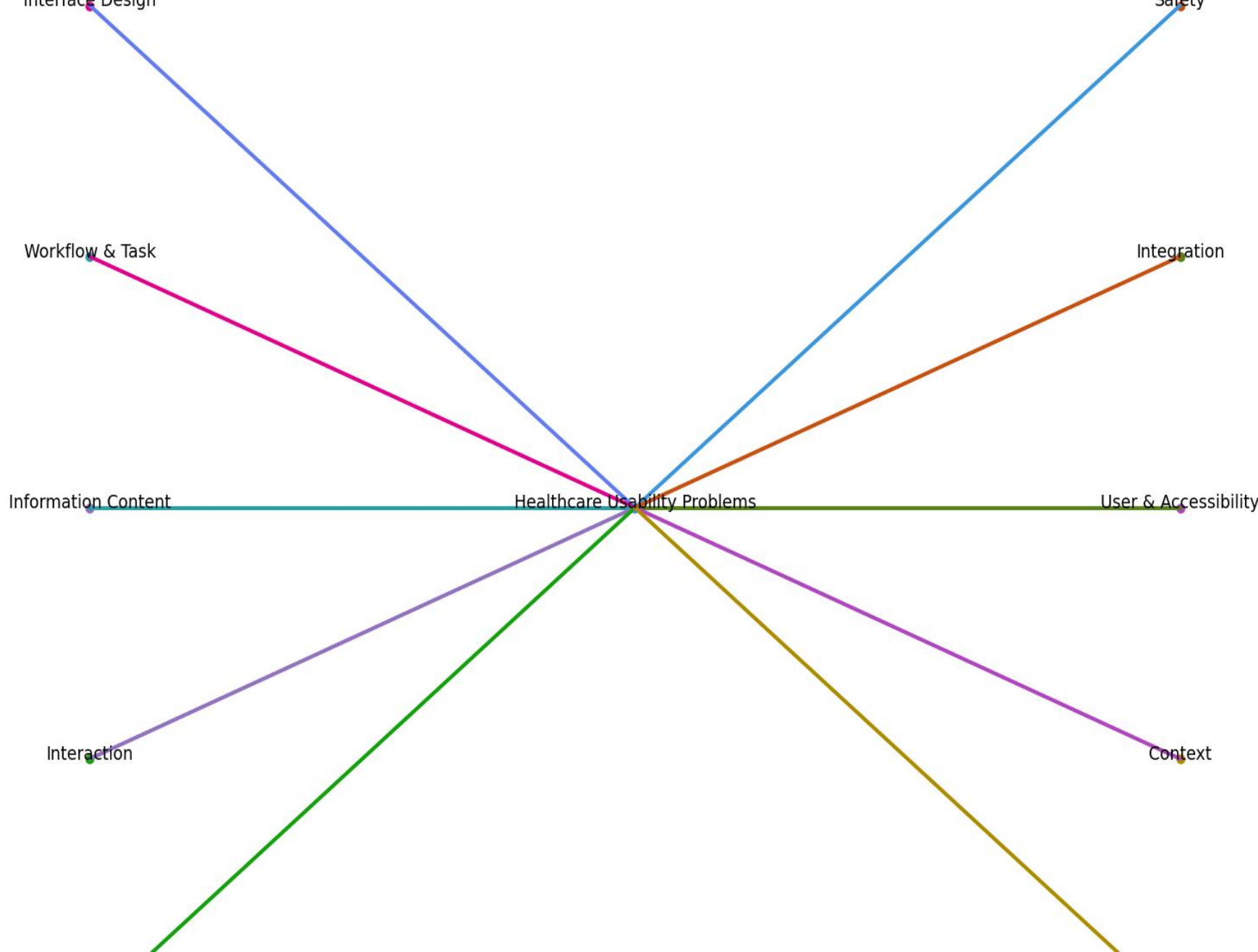


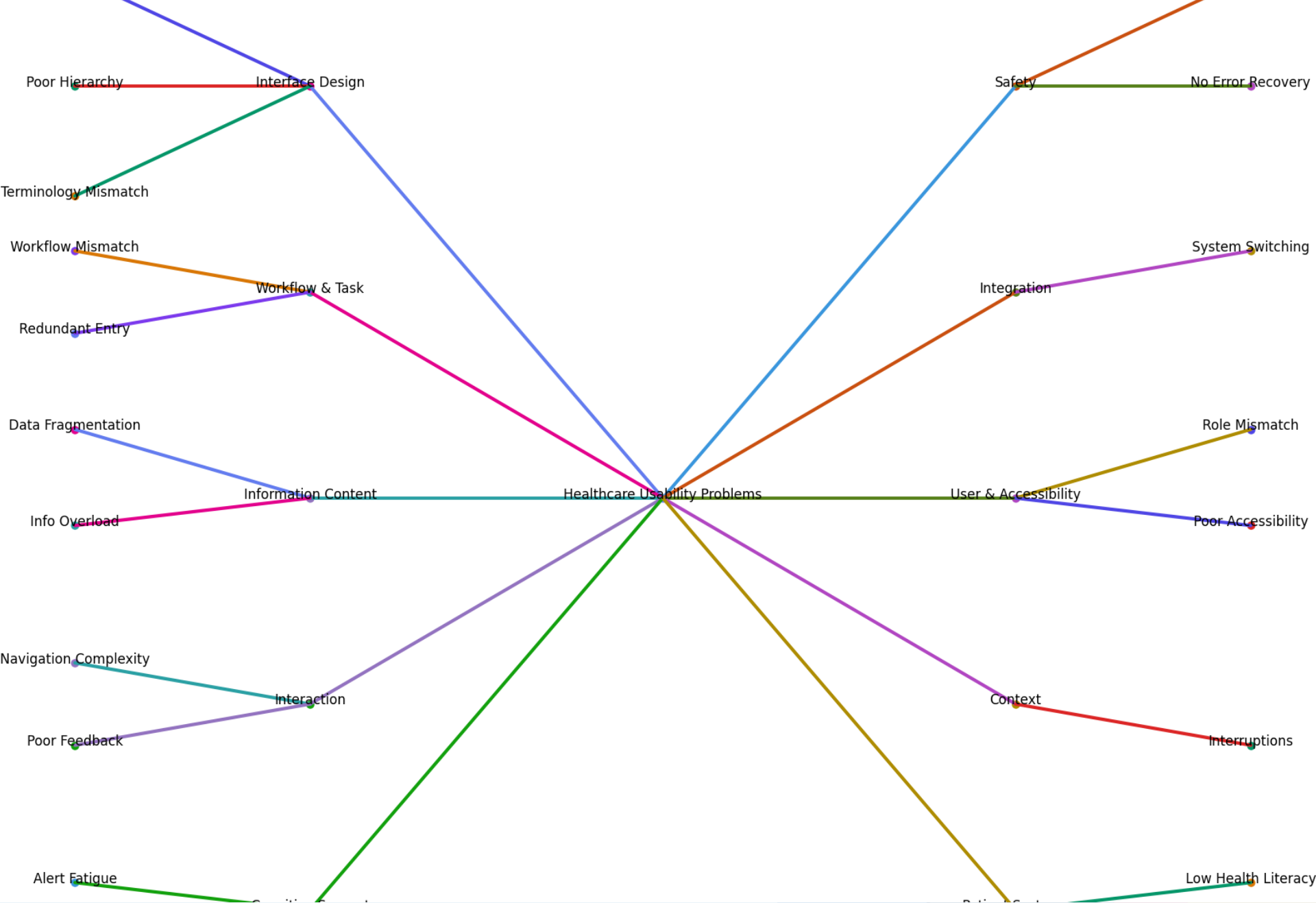
Reasoning about User Requirements for Health Devices (using i* modeling tool)

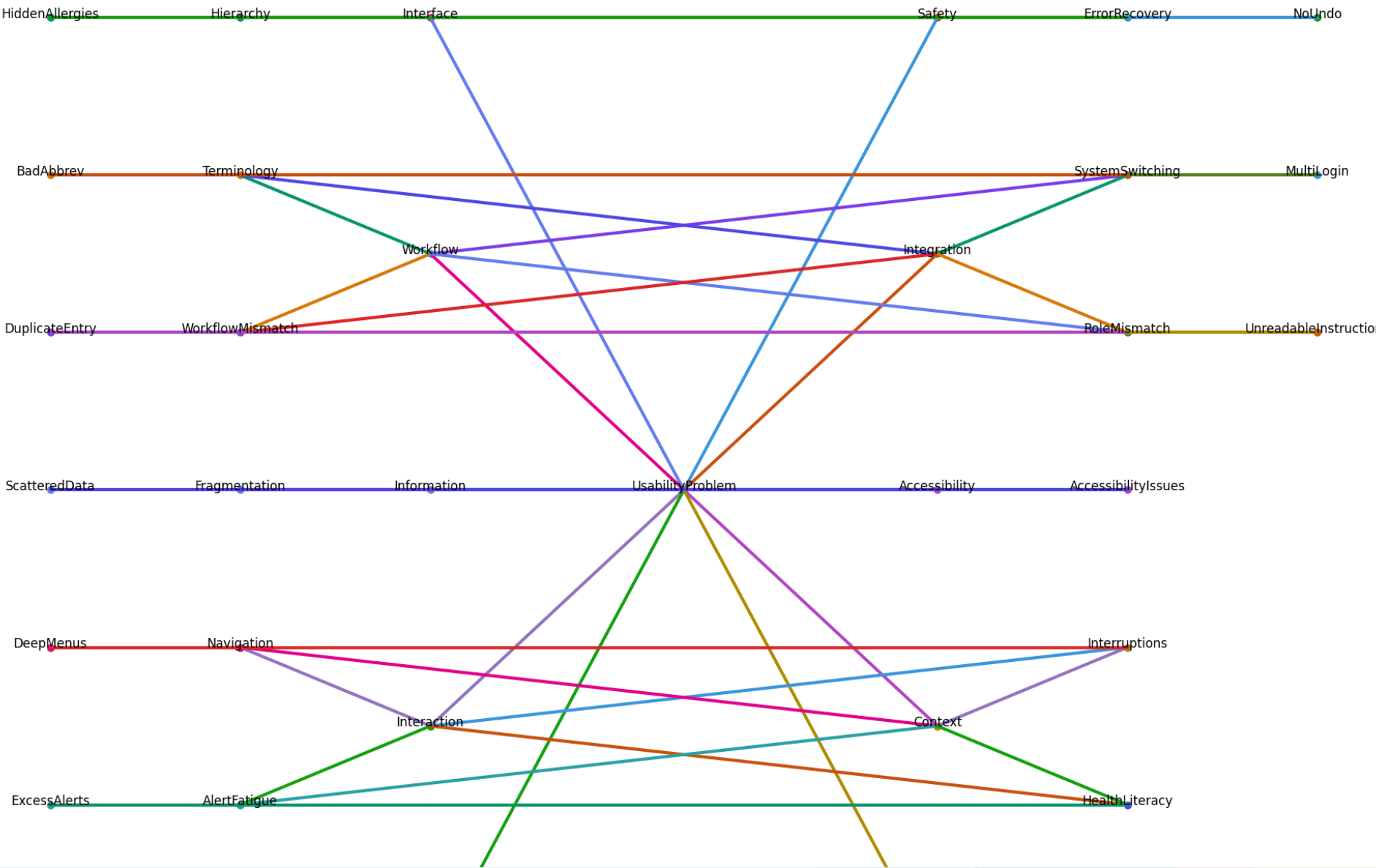


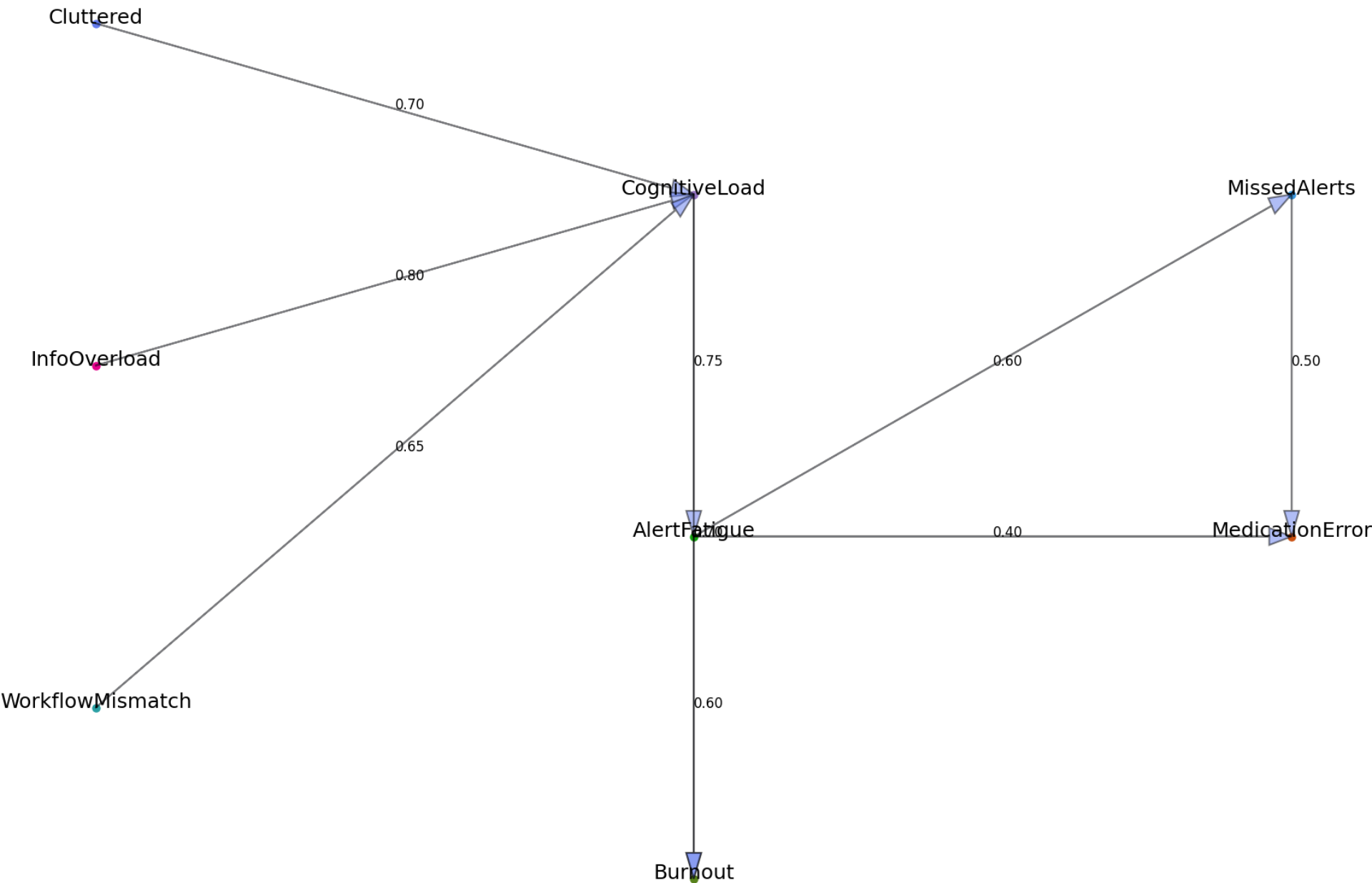
KNOWLEDGE GRAPHS FOR USABILITY AND HUMAN FACTORS RESEARCH – MY CURRENT RESEARCH

- Applying generative AI and LLMs to explore the relationship between **usability**, **usability issues**, and **causal relationships underlying usability issues**
 - Using prompting, drilling down layers to develop and query ontologies about usability and usability problems
- Also as data inputs using growing body of literature and reports on usability (commercial and academic)
- Explorations conducted as a series of iterative prompts to LLMs (copilot, NotebookLM)
- Issue of **validation** of automatically generated graphs, need for human validation











KNOWLEDGE GRAPHS AND NEURO-SYMBOLIC AI

- With advent of artificial neural nets (ANNs), although focus on representing symbolic knowledge remained, in the past two decades the focus moved to ANNs and deep neural networks (DNNs) - the current AI approach that has garnered most of the renewed interest in AI
 - However, inherent limitations of deep learning systems has now led to renewed interest in knowledge representations to augment LLMs, hence knowledge graph integration with LLMs
- My attention moved to health informatics, cognitive science and human factors, but the issue of how to represent knowledge has remained in all my research

RETURN TO SYMBOLIC KNOWLEDGE REPRESENTATION

- Limitations of LLMs and related architecture
 - limitations representing meaning, generic knowledge/understanding, context and the environment, ability to represent causal relations, abstract concepts
- Symbolic knowledge representation makes a come back!
- Move to **Neuro-Symbolic AI**.
 - Also corresponds to move to **Agentic AI**, which also realized need to include and expand “classic AI” approaches to reasoning and knowledge representation -- for AI to have real value for complex real-world tasks in medicine
- Combination as a hybrid approach, with in-depth layered symbolic knowledge representation as a foundational pillar

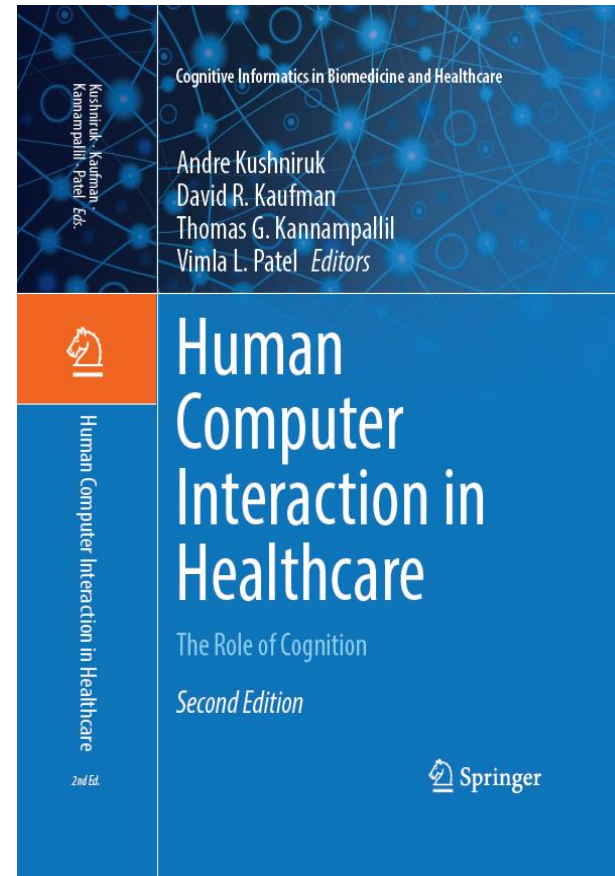
FINAL CONCLUSIONS

- AI research is concerned with the representation and implementation of human capabilities in a computer-based system.
- Although AI research and cognitive research are concerned with the processing of information, the latter is more concerned with understanding **human knowledge representation**
- However, there has been (and continues to be) a need for interdisciplinary research that blurs the boundary between **AI and cognitive science**
 - This will be needed in order to validate knowledge representations and graphs developed through automation
 - Will also be needed to serve as the core for more complex agentic AI

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